Cross-Domain Plant Disease Classification

Deep Learning for Visual Recognition

Problem

Models trained on lab images fail in the field due to domain shift. We close this gap.

Data

Train: PlantVillage (lab)
Test: PlantDoc (field)
26 shared disease classes

Approach

Evaluate 6 architectures and 13 training strategies.
Propose Style Disentanglement for domain-invariant learning.

Best Result

Macro F1 = 0.45 on PlantDoc (Swin-B + perspective aug, zero-shot transfer)

Impact

A decision-support tool that narrows 26 diseases to the top 3 with confidence guidance.

Repo Link: https://github.com/Jenpoer/plant-disease-detection.git

Problem Statement

- Plant diseases cause 20 – 40% crop losses worldwide (FAO)
- Deep learning enables automated visual diagnosis from leaf images
- Lab-trained models fail in the field — Can augmentation strategies and architecture choice bridge the lab-to-field generalization gap?

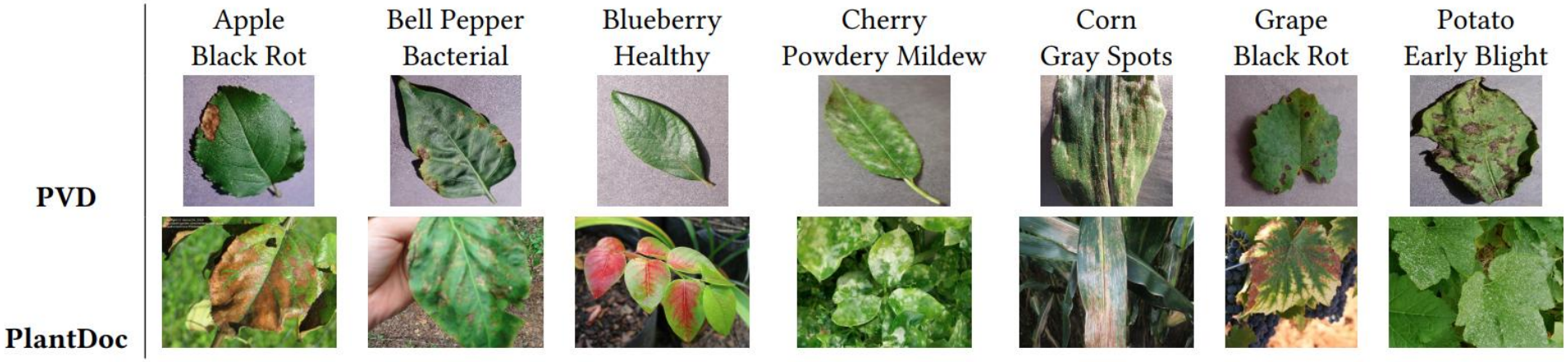


Figure 1: Samples from various classes in the PlantDoc Dataset show the gap between lab-controlled and real-life images

Datasets

26 shared disease classes mapped across both datasets

Dataset	Role	Source	Conditions
PlantVillage	Train / Val	Lab images	Clean, controlled
PlantDoc	Test (OOD)	Web-scraped	Field, noisy
PV_Test	In-domain test	Lab images	~99% F1 (saturated)

Baselines

For baselines, we **evaluated 6 different backbones** pre-trained on ImageNet with standard fine-tuning and with LoRA.

Architecture	Type	Baseline		Fine-tuned (LoRA)	
		PlantVillage F1	PlantDoc F1	PlantVillage F1	PlantDoc F1
MobileNetV3-Small	CNN	0.99	0.15	N/A	N/A
EfficientNet-B0	CNN	0.99	0.18	N/A	N/A
ViT-B/16	Transformer	0.98	0.30	0.98	0.33
Swin-B	Transformer	1.00	0.26	0.99	0.38
MaxViT-B	Transformer	0.99	0.29	N/A*	N/A*
CCT-14/7×2	Transformer	0.99	0.28	0.99	0.30

** Failed to run due to compute constraints*

Augmentation Strategies

Geometric

- Rotation
- Rotation + Gaussian Blur
- Affine

Mixing

- CutMix
- MixUp
- CutMixUp

Erasing

- Random Erasing
- Gaussian Blur

Perspective

- Zero-shot
- Finetuned

→ **Style Disentanglement**

- Total = 13 configurations × 6 architectures = **164 experimental runs**
- Style Disentanglement: **incorporates the different augmentation methods to try and learn domain-invariant features**

Key Results — Top 3 Augmentations by Backbone (PlantDoc)

Backbone	Rank	Best Method	Acc	Macro F1
CCT-14	**1**	**Random Erasing**	**0.39**	**0.35**
	2	Random Rotation	0.39	0.34
	3	Random Rotation + Gaussian Blur	0.37	0.33
EfficientNet-B0	**1**	**Perspective (zero-shot)**	**0.27**	**0.33**
	2	Perspective (fine-tuned)	0.27	0.24
	3	Gaussian Blur Only	0.24	0.23
MaxViT-B	**1**	**Perspective (fine-tuned)**	**0.35**	**0.32**
	2	CutMix	0.34	0.32
	3	Random Erasing	0.34	0.29
MobileNetV3-Small	**1**	**Style Disentanglement**	**0.24**	**0.21**
	2	Perspective (fine-tuned)	0.24	0.20
	3	Perspective (zero-shot)	0.23	0.18
Swin-B	**1**	**Perspective (zero-shot)**	**0.48**	**0.45**
	2	Style Disentanglement	0.47	0.42
	3	Baseline (LoRA)	0.43	0.38
ViT-B/16	**1**	**Random Erasing + AdamW**	**0.41**	**0.37**
	2	Style Disentanglement	0.38	0.34
	3	Baseline (LoRA)	0.37	0.33

Key Results — Best per Experiment (PlantDoc)

Rank	Experiment	Best Model	Acc	Macro F1
1	**Perspective (zero-shot)**	**Swin-B**	**0.48**	**0.45**
2	Style Disentanglement	Swin-B	0.47	0.42
3	Baseline (LoRA)	Swin-B	0.43	0.38
4	Random Erasing (+AdamW)	ViT-B/16	0.41	0.37
5	Random Erasing	CCT-14	0.39	0.35
6	Rotation Only	CCT-14	0.39	0.34
7	Gaussian Blur	Swin-B	0.36	0.33
8	Affine	Swin-B	0.37	0.32
9	CutMix	Swin-B	0.37	0.32
10	Perspective (fine-tuned)	CCT-14	0.37	0.32
11	CutMixUp	Swin-B	0.34	0.32
12	Baseline	MaxViT-B	0.33	0.29
13	MixUp	Swin-B	0.33	0.28

Swin-B + Perspective (zero-shot) achieves the highest PlantDoc Macro F1 of 0.45

Perspective Augmentation — Deep Dive

Comparison between fine-tuning and without fine-tuning on Swin-B

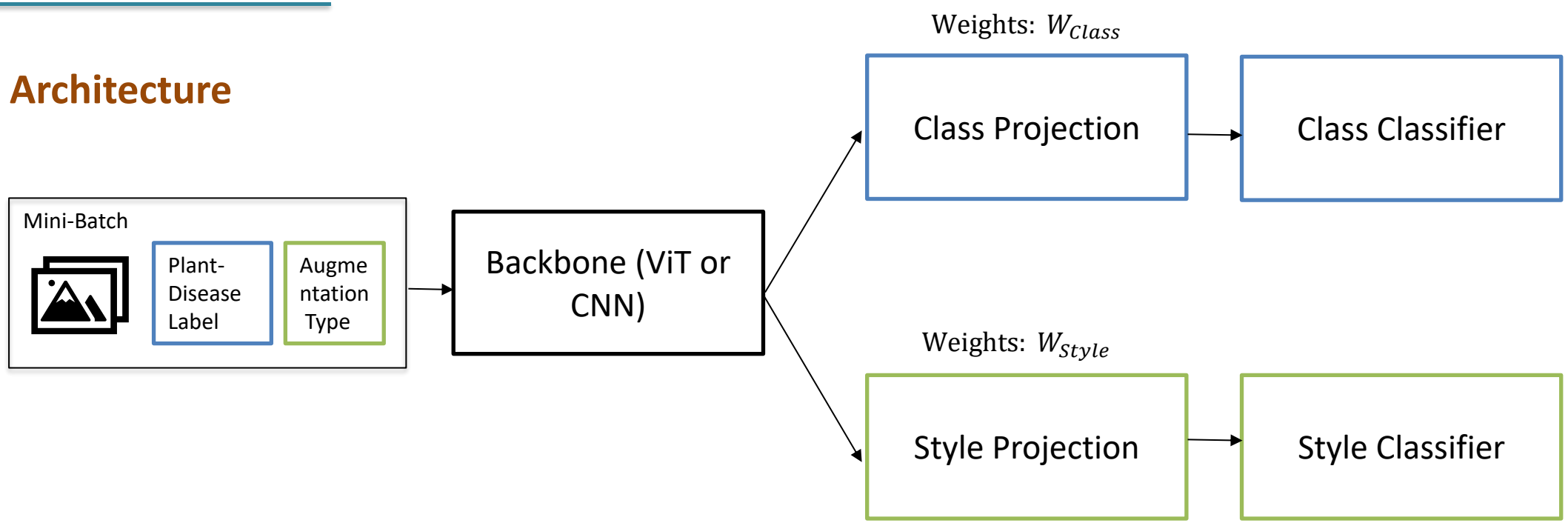
Variant	PlantDoc Acc	Macro F1
Zero-shot (frozen backbone)	**0.48**	**0.45**
Finetuned backbone	0.33	0.28

Key Insight :

- **Frozen backbone preserves ImageNet features** that generalize to PlantDoc
- **Fine-tuning overwrites these features, hurting cross-domain transfer**
- Perspective distortion **simulates real-world viewing angles → reduces domain gap**
- Zero-shot outperforms finetuned by +0.17 Macro F1

Style Disentanglement

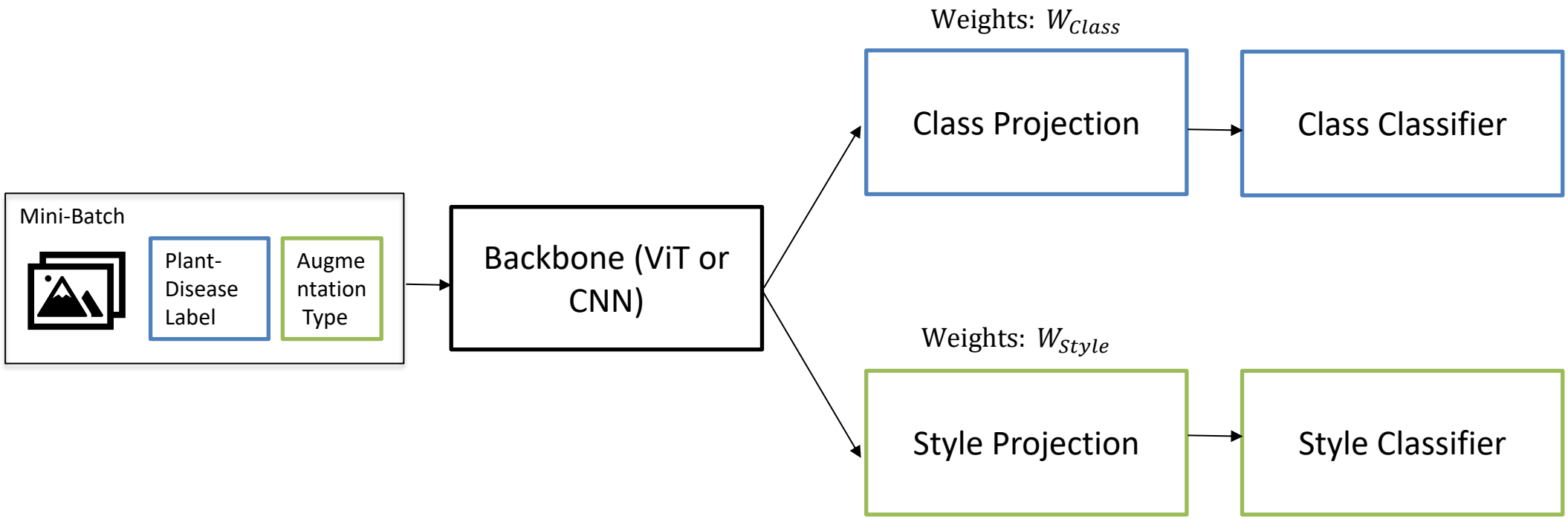
Architecture



$$L = L_{CE}(Class) + L_{CE}(Style) + L_{Supcon*}(Class) + L_{Supcon*}(Style) + L_{Orth}(W_{Class}, W_{Style})$$

Main Idea: Have the model **project the latent features from the backbone into different subspaces** for class & "style" (i.e. augmentations in the image)

** Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., & Krishnan, D. (2020). Supervised contrastive learning. In Advances in Neural Information Processing Systems (Vol. 33). https://arxiv.org/abs/2004.11362.*



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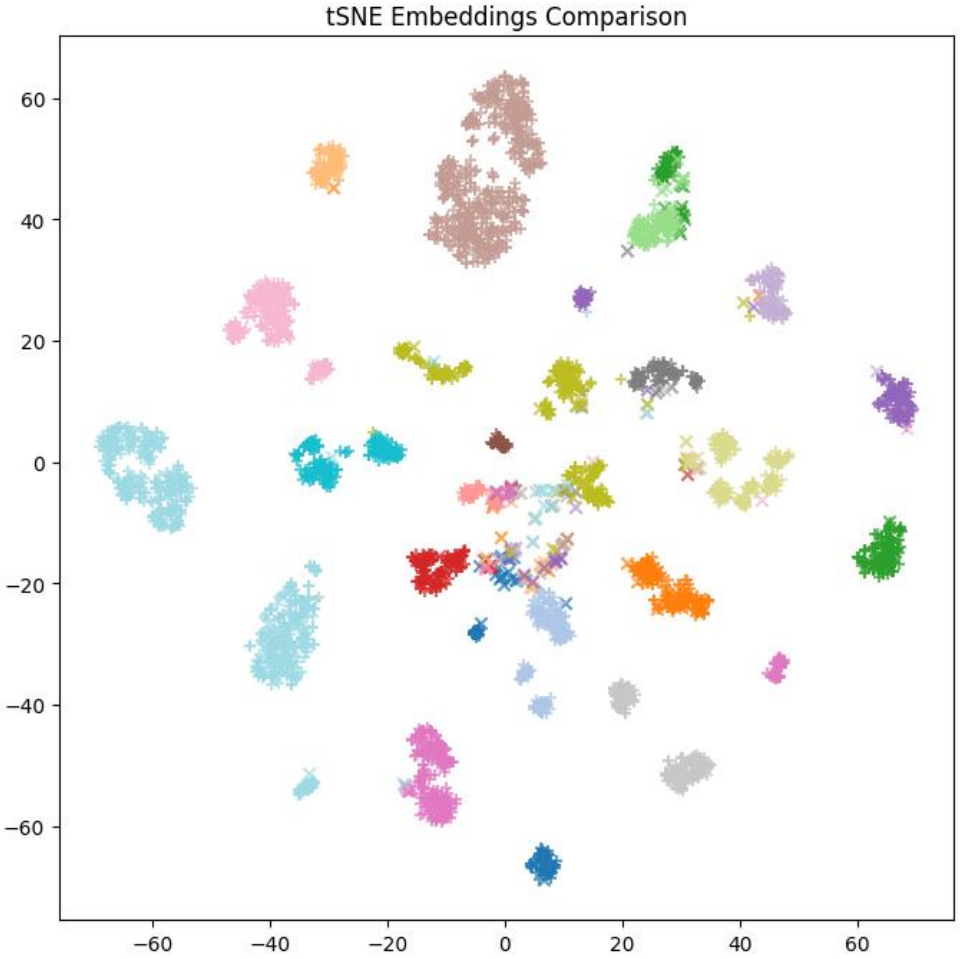
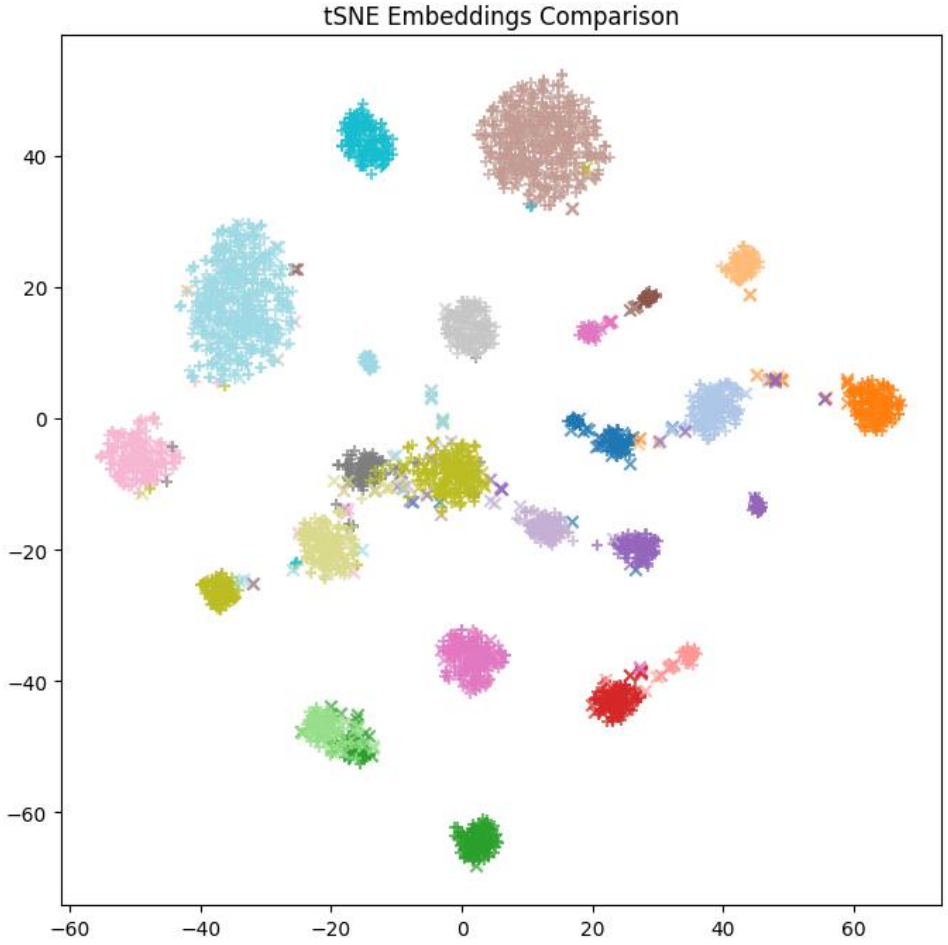
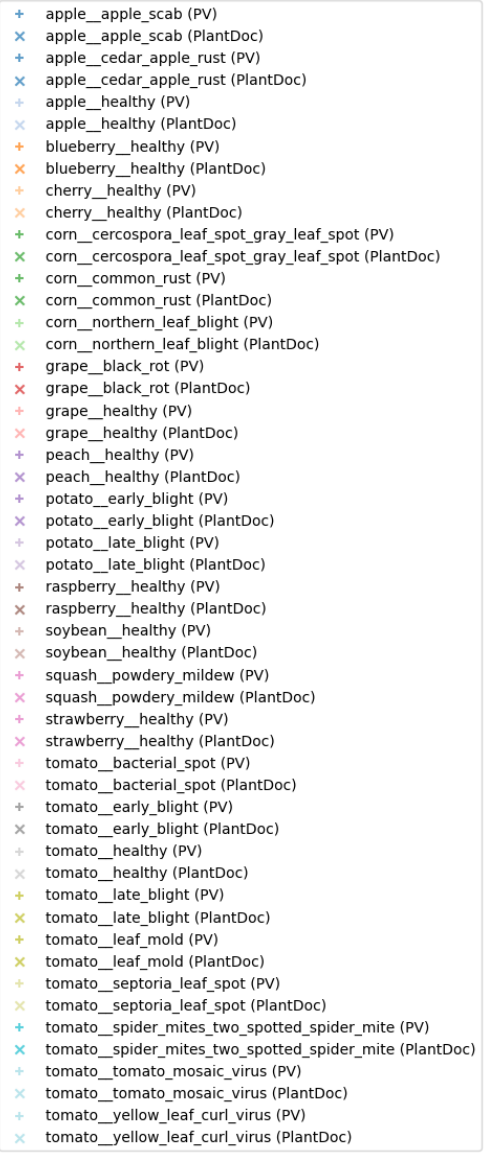
Style Disentanglement – Embeddings Analysis

Baseline (Swin-B)



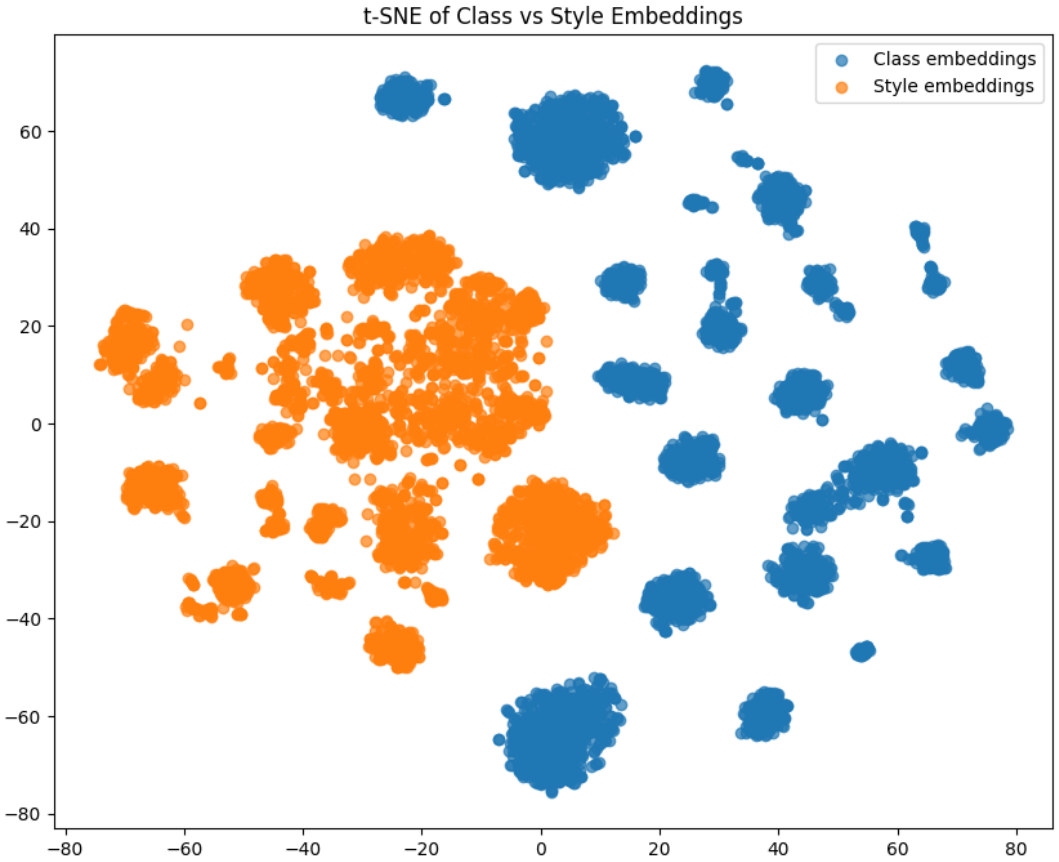
Style-Disentangled (Swin-B)





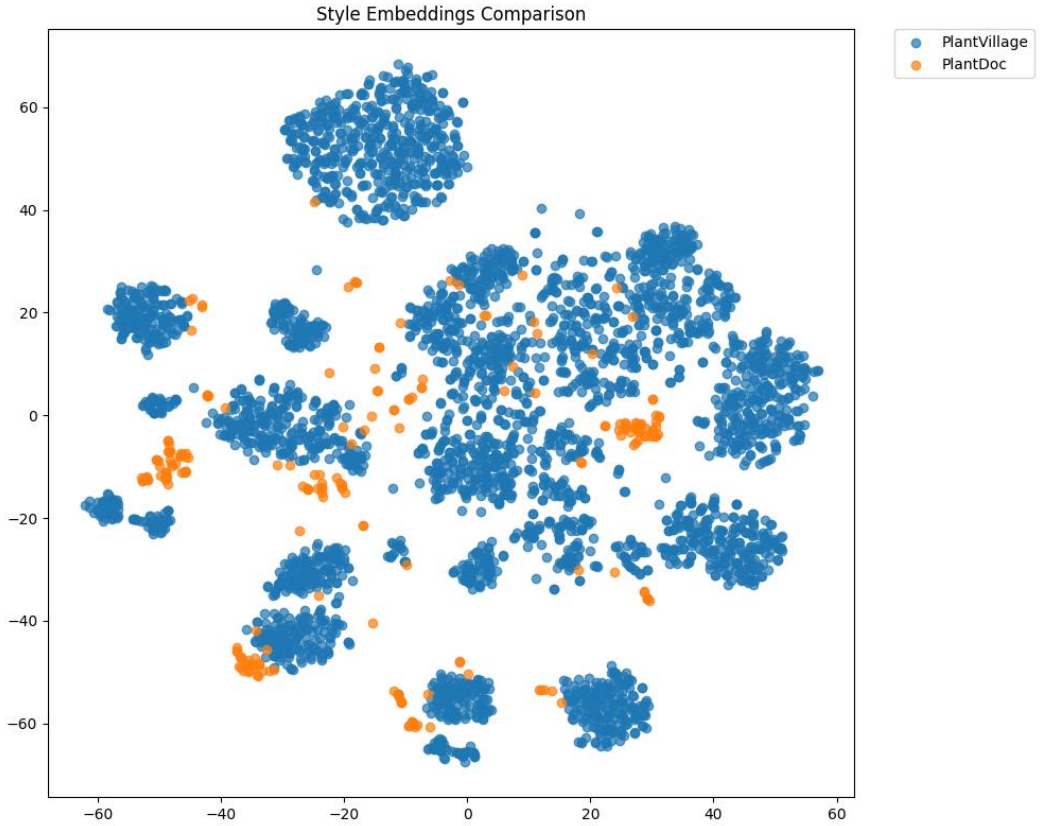
Style Disentanglement – Embeddings Analysis

Comparison of Projections



Style and Class Features are disentangled

Style Embeddings



However, our Style Projection is **still not good enough to cleanly distinguish** PlantDoc and PlantVillage

Comparison with Prior Work

- Our best result (0.45 for Macro F1) **surpasses the prior benchmark (0.38)**
- **Achieved with a simple augmentation strategy + frozen backbone**

+18.4%
relative improvement in Macro F1

Study	Method	PlantDoc Macro F1
Singh et al. (baseline ref.)	Standard fine-tuning	0.38
Our Current Best Model	**Swin-B + Perspective (zero-shot)**	**0.45**

Computer Science > Computer Vision and Pattern Recognition

[Submitted on 23 Nov 2019]

PlantDoc: A Dataset for Visual Plant Disease Detection

Davinder Singh, Naman Jain, Pranjali Jain, Pratik Kayal, Sudhakar Kumawat, Nipun Batra

India loses 35% of the annual crop yield due to plant diseases. Early detection of plant diseases remains difficult due to the lack of lab infrastructure and expertise. In this paper, we explore the possibility of computer vision approaches for scalable and early plant disease detection. The lack of availability of sufficiently large-scale non-lab data set remains a major challenge for enabling vision based plant disease detection. Against this background, we present PlantDoc: a dataset for visual plant disease detection. Our dataset contains 2,598 data points in total across 13 plant species and up to 17 classes of diseases, involving approximately 300 human hours of effort in annotating internet scraped images. To show the efficacy of our dataset, we learn 3 models for the task of plant disease classification. Our results show that modelling using our dataset can increase the classification accuracy by up to 31%. We believe that our dataset can help reduce the entry barrier of computer vision techniques in plant disease detection.

Key Takeaways

1. **Perspective augmentation + frozen backbone achieves the best cross-domain transfer** (Macro F1: 0.45)

2. Style disentanglement (Macro F1: 0.42) and LoRA (Macro F1: 0.38) are strong alternatives — **domain generalization methods complement augmentation**

3. **Vision Transformers (Swin-B, CCT-14) consistently outperform CNNs**; Swin-B is the best model in 8 of 13 experiments

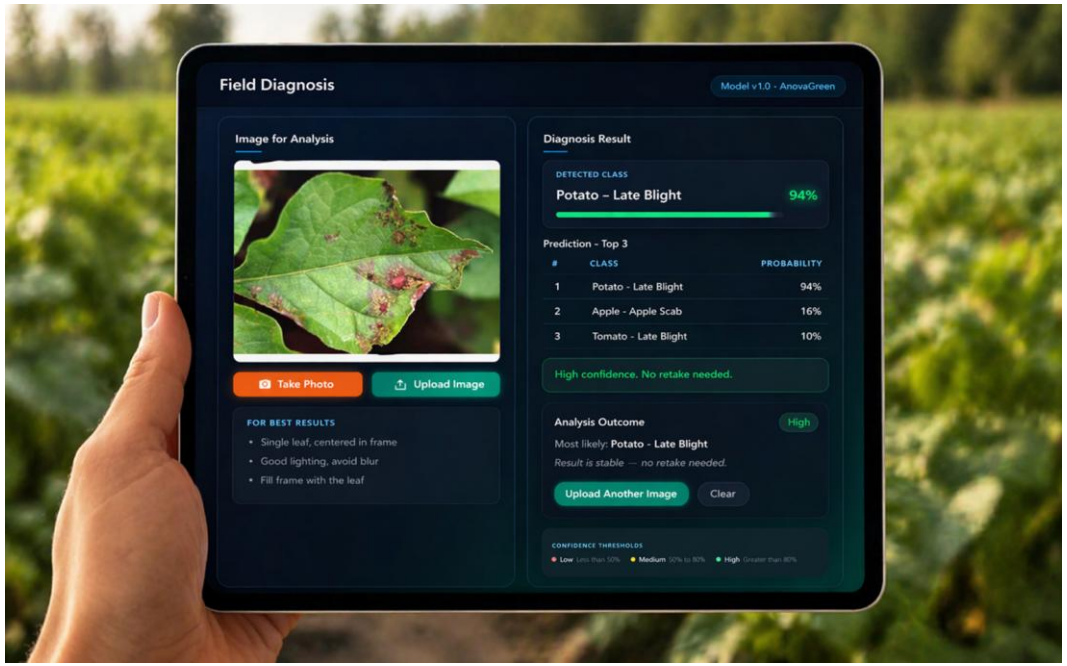
Limitations

- Single hyperparameter configuration per model
- No augmentation combinations tested
- 10-epoch budget may limit convergence
- No confidence calibration or reliability analysis

Future Work

- Combine top augmentations (erasing + perspective)
- Come up with better augmentation techniques (e.g. splicing leaf images with non-leaf artifacts (e.g. images of human hand, fruits, etc); ROI cropping)
- Longer training: 25–30 epochs + cosine LR
- Explicit domain adaptation

Future Application: Field Diagnosis



Decision-support tool, not a replacement — model shortlists, farmer decides

How It Works

- Upload a leaf photo or use live camera capture
- Returns top-3 predictions with confidence scores
- Confidence bands (Low / Medium / High) guide decisions
- Supports 26 disease classes across 10 plant types

Why Top-3?

- At 0.45 Macro F1, the model is not accurate enough to trust a single prediction
- Top-3 narrows 26 diseases → 3 likely candidates for the farmer to judge
- Farmer applies local knowledge (crop type, season, region) to make the final call

Tech: FastAPI + React + ViT-B/16 (CPU inference)